



Machine Learning



Module 3

Linear Models

Least Square Method

- The Least Squares Method is used to derive a generalized linear equation between two variables, one of which is independent and the other dependent on the former.
- The value of the independent variable is represented as the x-coordinate and that of the dependent variable is represented as the y-coordinate in a 2D cartesian coordinate system.
- The least-squares method can be defined as a statistical method that is used to find the equation of the line of best fit related to the given data. This method is called so as it aims at reducing the sum of squares of deviations as much as possible. The line obtained from such a method is called a **regression line**.

Least Square Method

- The steps used in deriving the line of best fit from this method are as follows:
 - **Step 1:** Denote the independent variable values as x_i and the dependent ones as y_i .
 - **Step 2:** Calculate the average values of x_i and y_i as X and Y .
 - **Step 3:** Presume the equation of the line of best fit as $y = mx + c$, where m is the slope of the line and c represents the intercept of the line on the Y-axis.
 - **Step 4:** The slope m can be calculated from the following formula:

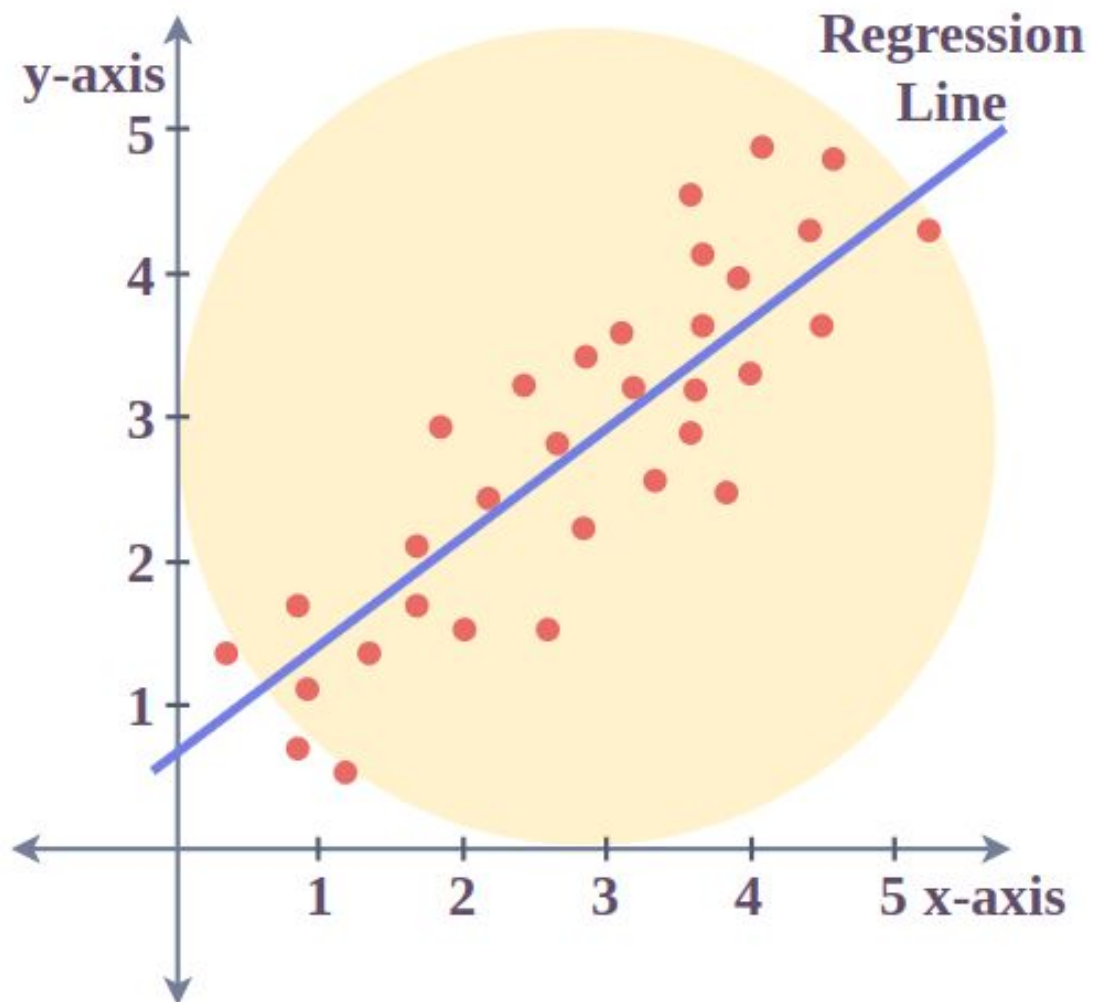
$$m = [\Sigma (X - x_i) \times (Y - y_i)] / \Sigma (X - x_i)^2$$

- **Step 5:** The intercept c is calculated from the following formula:

$$c = Y - mX$$

Least Square Method

- Least Square Method Graph



Least Square Method

Example:

Find the line of best fit for the following data points using the least squares method: $(x,y) = (1,3), (2,4), (4,8), (6,10), (8,15)$.

Solution:

Here, we have x as the independent variable and y as the dependent variable. First, we calculate the means of x and y values denoted by X and Y respectively.

$$X = (1+2+4+6+8)/5 = 4.2$$

$$Y = (3+4+8+10+15)/5 = 8$$

x_i	y_i	$X - x_i$	$Y - y_i$	$(X - x_i) * (Y - y_i)$	$(X - x_i)^2$
1	3	3.2	5	16	10.24
2	4	2.2	4	8.8	4.84
4	8	0.2	0	0	0.04
6	10	-1.8	-2	3.6	3.24
8	15	-3.8	-7	26.6	14.44
Sum (Σ)		0	0	55	32.8

- The slope of the line of best fit can be calculated from the formula as follows:

$$m = (\sum (X - x_i) * (Y - y_i)) / \sum (X - x_i)^2$$

$$m = 55/32.8 = 1.68 \text{ (rounded upto 2 decimal places)}$$

- Now, the intercept will be calculated from the formula as follows:

$$c = Y - mX$$

$$c = 8 - 1.68 * 4.2 = 0.94$$

- Thus, the equation of the line of best fit becomes,
 $y = 1.68x + 0.94$.

SIMPLE LINEAR REGRESSION

- shows the linear relationship between the independent variable (X-axis) and the dependent variable (Y-axis)
- **an algorithm that provides a linear relationship between an independent variable and a dependent variable to predict the outcome of future events.**
- A sloped straight line represents the linear regression model.

Multiple Linear Regression (MLR)

- predicts the value of a dependent variable (sometimes it is called as the outcome, target or criterion variable) on the basis of two or more independent variables (or sometimes, the predictor, explanatory or regressor variables)
- Examples:
 1. MLR can be used for predicting exam performance on the basis of revision time, lecture attendance, gender, or time anxiety.
 2. identifying the relationship between the distance covered (dependent variable) by the cab driver and the age of the driver and years of experience (independent variables).

Logistic regression

- comes under the Supervised Learning technique.
- used for predicting the categorical dependent variable using a given set of independent variables.
- predicts the output of a categorical dependent variable. Therefore the outcome must be a categorical or discrete value.
- It can be either Yes or No, 0 or 1, true or False, etc. but instead of giving the exact value as 0 and 1, **it gives the probabilistic values which lie between 0 and 1.**
- In Logistic regression, instead of fitting a regression line, we fit an "S" shaped logistic function, which predicts two maximum values (0 or 1).

Regularization in Machine Learning

- It is a technique to prevent the model from overfitting by adding extra information to it.
- Sometimes the machine learning model performs well with the training data but does not perform well with the test data. It means the model is not able to predict the output when deals with unseen data by introducing noise in the output, and hence the model is called overfitted. This problem can be deal with the help of a regularization technique.
- This technique can be used in such a way that it will allow to maintain all variables or features in the model by reducing the magnitude of the variables. Hence, it maintains accuracy as well as a generalization of the model.
- It mainly regularizes or reduces the coefficient of features toward zero. In simple words, *"In regularization technique, we reduce the magnitude of the features by keeping the same number of features."*

How does Regularization Work?

- Let's consider the simple linear regression equation:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \dots + \beta_n x_n + b$$

In the above equation,

Y represents the value to be predicted

X1, X2, ...Xn are the features for Y.

$\beta_0, \beta_1, \dots, \beta_n$ are the weights or magnitude attached to the features, respectively.

- Linear regression models try to optimize the β_0 and b to minimize the cost function.
- The equation for the cost function for the linear model is given below:

$$\sum_{i=1}^M (y_i - y'_i)^2 = \sum_{i=1}^M (y_i - \sum_{j=0}^n \beta_j * X_{ij})^2$$

that can predict the accurate value of Y. The loss function for the linear regression is called as **RSS or Residual sum of squares**.

Techniques of Regularization

There are mainly two types of regularization techniques:

- 1. Ridge Regression(L2 regularization.)**
- 2. Lasso Regression(L1 regularization)**

Techniques of Regularization

1. Ridge Regression

- Ridge regression is one of the types of linear regression in which a small amount of bias is introduced so that we can get better long-term predictions.
- Ridge regression is a regularization technique, which is used to reduce the complexity of the model. It is also called as **L2 regularization**.
- In this technique, the cost function is altered by adding the penalty term to it. The amount of bias added to the model is called **Ridge Regression penalty**. We can calculate it by multiplying with the lambda to the squared weight of each individual feature.
- The equation for the cost function in ridge regression will be:

$$\sum_{i=1}^M (y_i - y'_i)^2 = \sum_{i=1}^M \left(y_i - \sum_{j=0}^n \beta_j * x_{ij} \right)^2 + \lambda \sum_{j=0}^n \beta_j^2$$

Techniques of Regularization

2. Lasso Regression:

- Lasso regression is another regularization technique to reduce the complexity of the model. It stands for **Least Absolute and Selection Operator**.
- It is similar to the Ridge Regression except that the penalty term contains only the absolute weights instead of a square of weights.
- Since it takes absolute values, hence, it can shrink the slope to 0, whereas Ridge Regression can only shrink it near to 0.
- It is also called as **L1 regularization**. The equation for the cost function of Lasso regression will be:

$$\sum_{i=1}^M (y_i - y'_i)^2 = \sum_{i=1}^M \left(y_i - \sum_{j=0}^n \beta_j * x_{ij} \right)^2 + \lambda \sum_{j=0}^n |\beta_j|$$

Techniques of Regularization

2. Lasso Regression:

- Some of the features in this technique are completely neglected for model evaluation.
- Hence, the Lasso regression can help us to reduce the overfitting in the model as well as the feature selection.

■ Key Difference between Ridge Regression and Lasso Regression

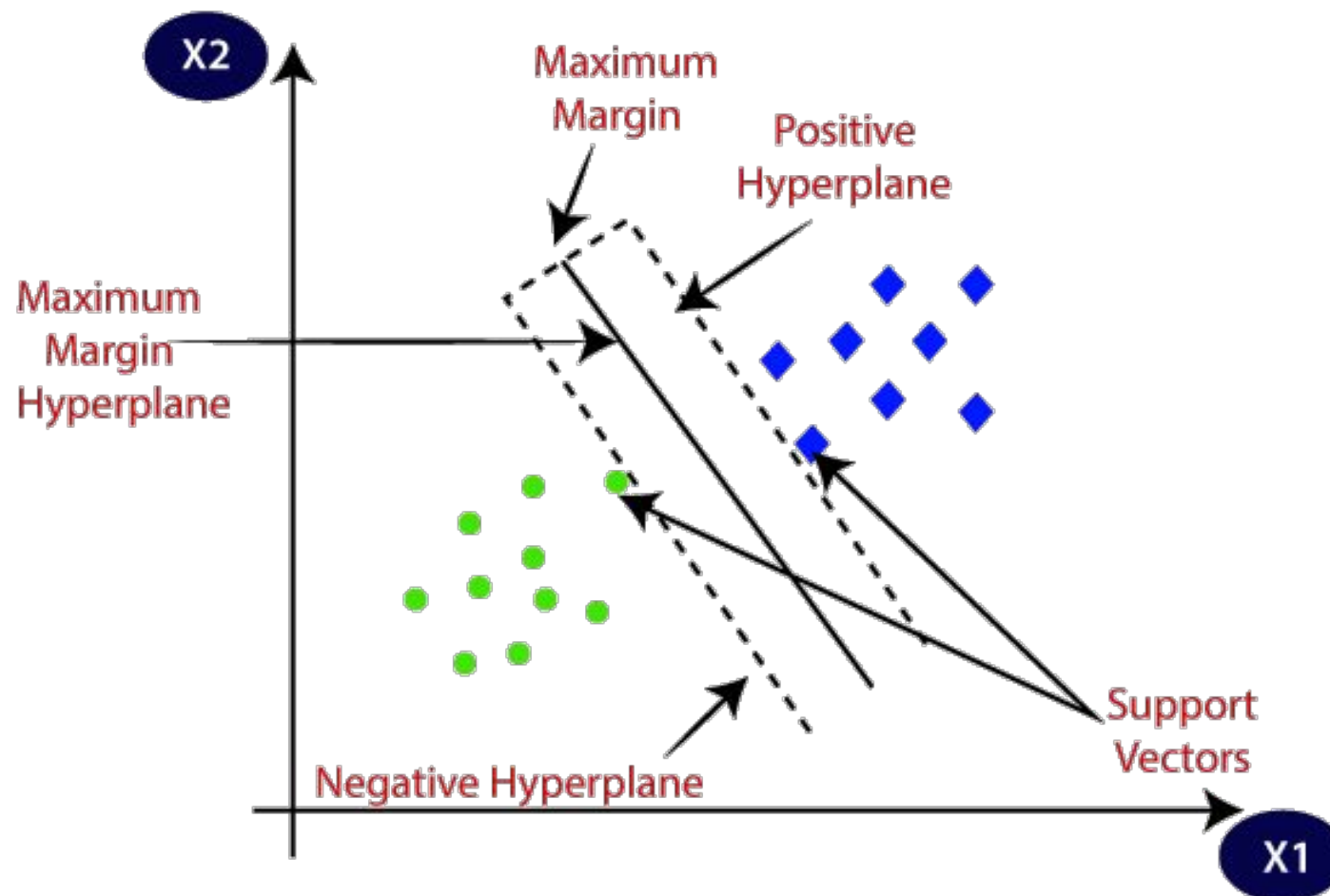
- **Ridge regression** is mostly used to reduce the overfitting in the model, and it includes all the features present in the model. It reduces the complexity of the model by shrinking the coefficients.
- **Lasso regression** helps to reduce the overfitting in the model as well as feature selection.

Support Vector Machine

- Support Vector Machine or SVM is one of the most popular Supervised Learning algorithms, which is used for Classification as well as Regression problems.
- However, primarily, it is used for Classification problems in Machine Learning.
- The goal of the SVM algorithm is to create the best line or decision boundary that can segregate n-dimensional space into classes so that we can easily put the new data point in the correct category in the future.
- This best decision boundary is called a hyperplane.
- SVM chooses the extreme points/vectors that help in creating the hyperplane.
- These extreme cases are called as support vectors, and hence algorithm is termed as Support Vector Machine.

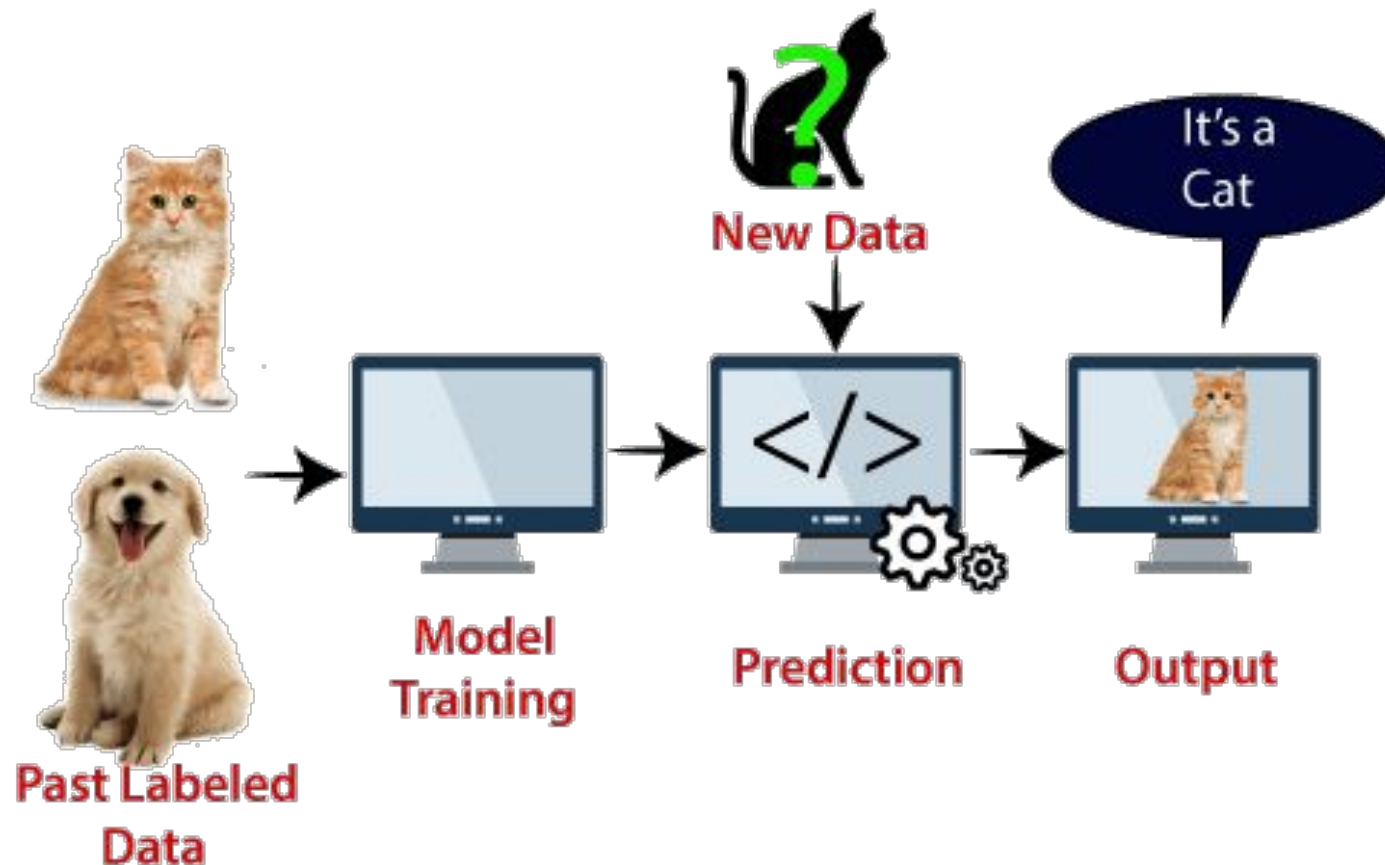
Support Vector Machine

- Consider the below diagram in which there are two different categories that are classified using a decision boundary or hyperplane:



Support Vector Machine

- **Example:** Suppose we see a strange cat that also has some features of dogs, so if we want a model that can accurately identify whether it is a cat or dog, so such a model can be created by using the SVM algorithm.



Types of SVM

- **SVM can be of two types:**
 1. **Linear SVM:** Linear SVM is used for linearly separable data, which means if a dataset can be classified into two classes by using a single straight line, then such data is termed as linearly separable data, and classifier is used called as Linear SVM classifier.
 2. **Non-linear SVM:** Non-Linear SVM is used for non-linearly separated data, which means if a dataset cannot be classified by using a straight line, then such data is termed as non-linear data and classifier used is called as Non-linear SVM classifier.

Hyperplane and Support Vectors in the SVM algorithm

■ **Hyperplane:**

- There can be multiple lines/decision boundaries to segregate the classes in n -dimensional space, but we need to find out the best decision boundary that helps to classify the data points. This best boundary is known as the hyperplane of SVM.
- The dimensions of the hyperplane depend on the features present in the dataset, which means if there are 2 features, then hyperplane will be a straight line. And if there are 3 features, then hyperplane will be a 2-dimension plane.

Hyperplane and Support Vectors in the SVM algorithm

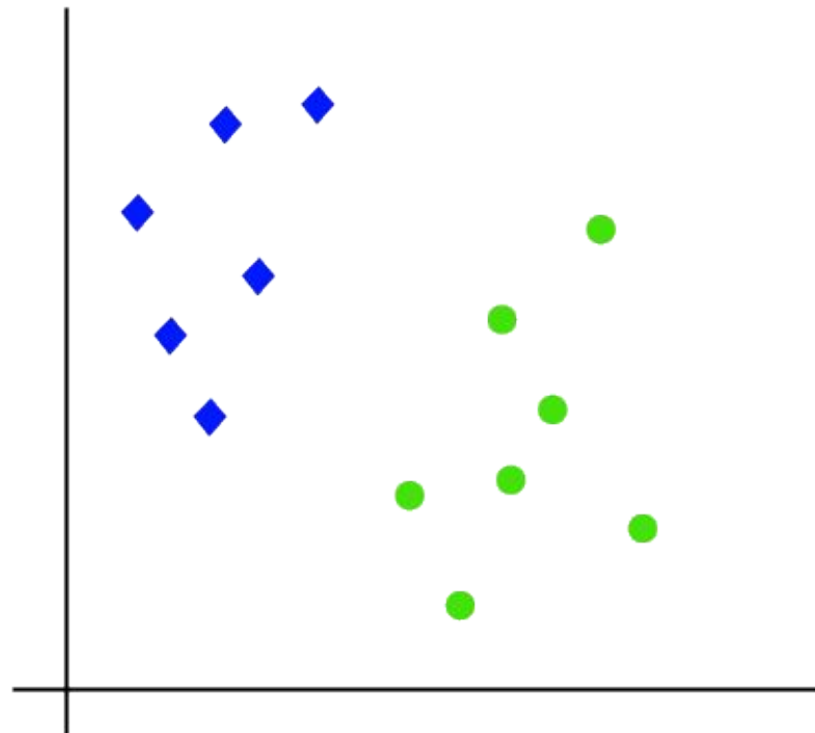
■ **Support Vectors:**

- The data points or vectors that are the closest to the hyperplane and which affect the position of the hyperplane are termed as Support Vector.
- Since these vectors support the hyperplane, hence called a Support vector.

How does SVM works?

■ Linear SVM:

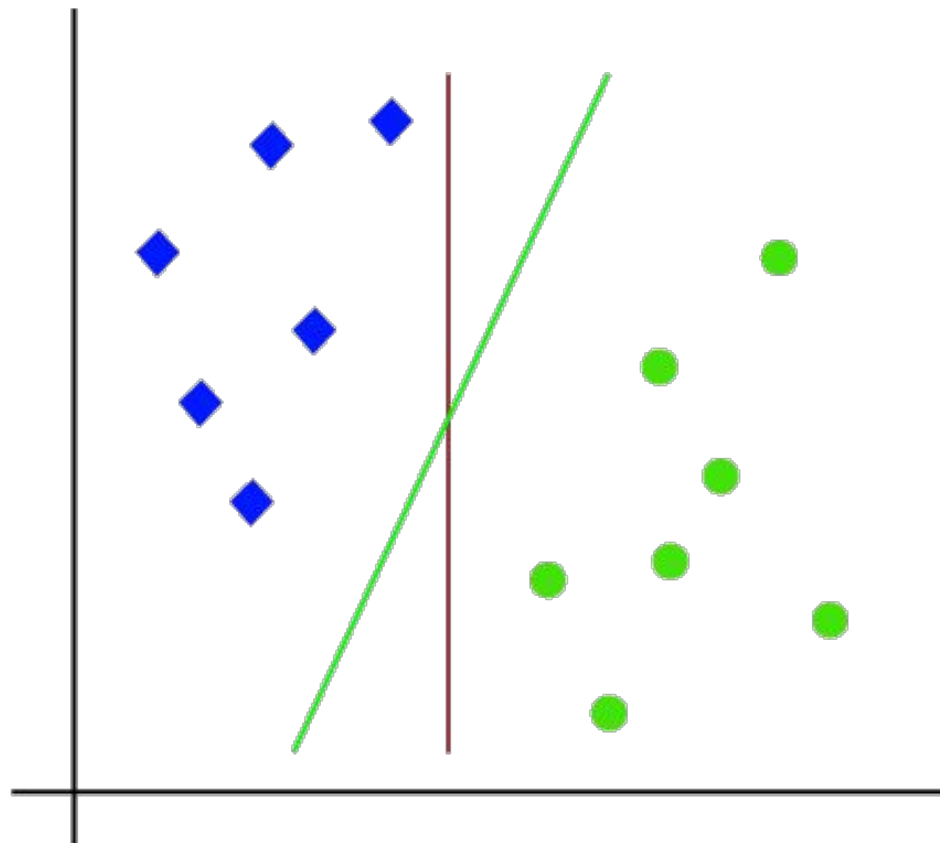
- The working of the SVM algorithm can be understood by using an example.
- Suppose we have a dataset that has two tags (green and blue), and the dataset has two features x_1 and x_2 .
- We want a classifier that can classify the pair(x_1 , x_2) of coordinates in either green or blue. Consider the below image:.



How does SVM works?

- **Linear SVM:**

- So as it is 2-d space so by just using a straight line, we can easily separate these two classes.
- But there can be multiple lines that can separate these classes. Consider the below image:

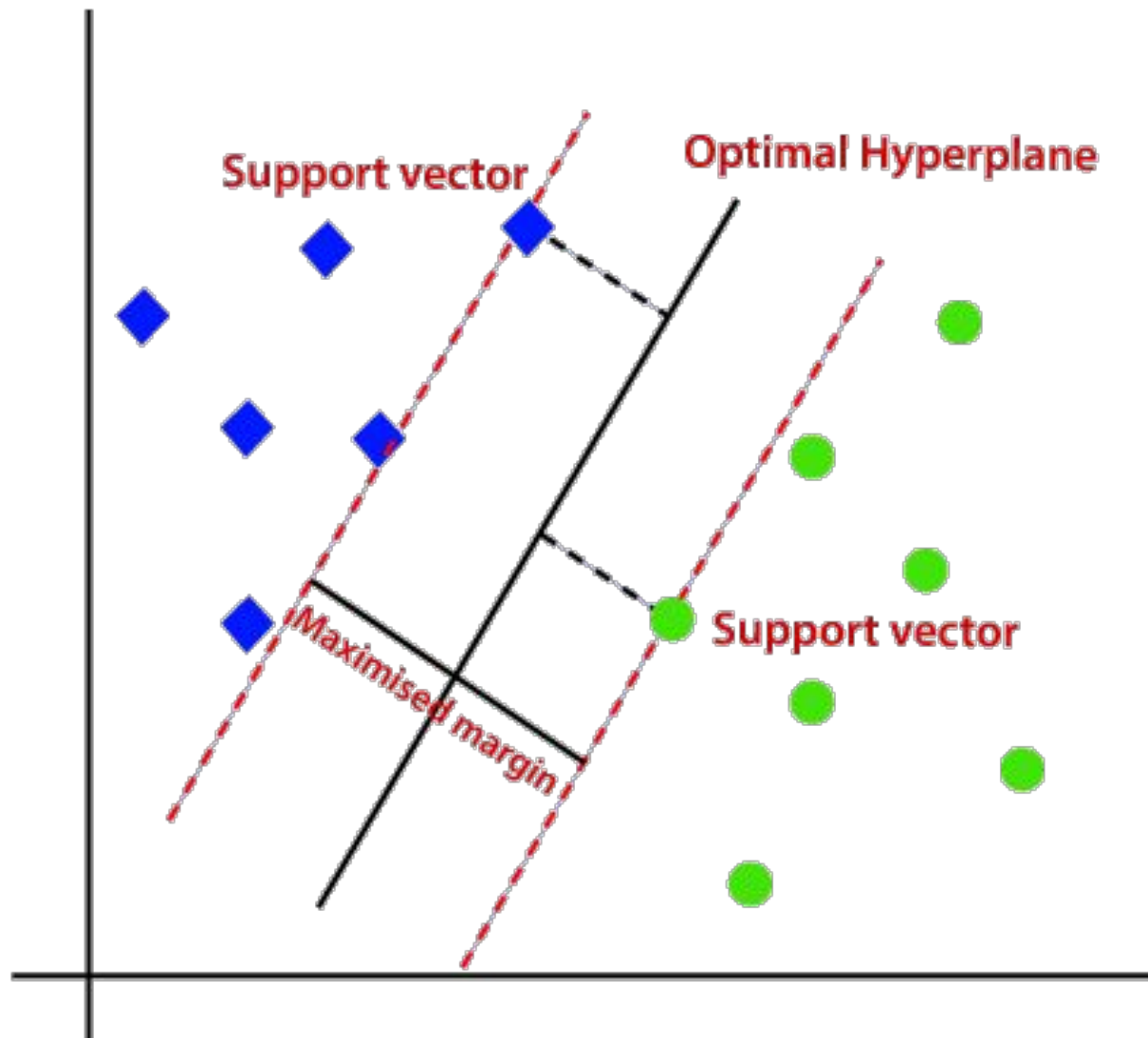


How does SVM works?

■ **Linear SVM:**

- Hence, the SVM algorithm helps to find the best line or decision boundary; this best boundary or region is called as a hyperplane.
- SVM algorithm finds the closest point of the lines from both the classes.
- These points are called support vectors.
- The distance between the vectors and the hyperplane is called as margin.
- And the goal of SVM is to maximize this margin.
- The hyperplane with maximum margin is called the optimal hyperplane.

How does SVM works?



How does SVM works?

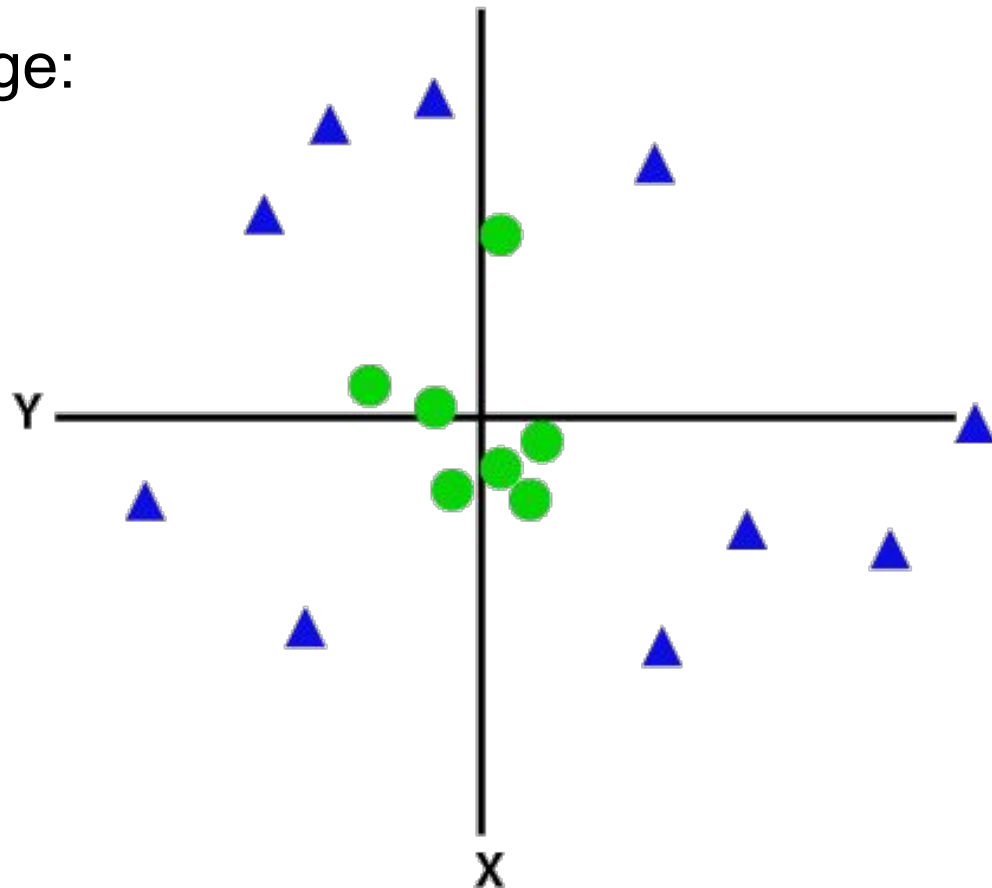
■ **Linear SVM:**

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How does SVM works?

■ Non-Linear SVM:

- If data is linearly arranged, then we can separate it by using a straight line, but for non-linear data, we cannot draw a single straight line.
- Consider the below image:



How does SVM works?

- **Non-Linear SVM:**

- **So to separate these data points, we need to add one more dimension.**
- **For linear data, we have used two dimensions x and y, so for non-linear data, we will add a third dimension z.**
- **It can be calculated as:**

$$z=x^2 +y^2$$

How does SVM works?

- **Non-Linear SVM:**
- **SVM will divide the datasets into classes in the following way.**

